

```

00001      " Name:ABS Origin Search
00002      " Func:ABS Origin Serach Control
00003      " From:MPM011
00004
00005      VAR;
00006 00000          LONG Axis_IDX1 = 0          " Axis Index1
00007 00001          LONG Axis_IDX2 = 0          " Axis Index2
00008 00002          LONG Axis_IDX3 = 0          " Axis Index3
00009 00003          LONG L1_PW = 0,            " Linel Axis Word Offset
00010 00004          LONG L1_PL = 0,            " Linel Axis Word Offset
00011
00012 00005          LONG Axis_IO_Chuck1 = 0;
00013 00006          LONG Axis_IO_B_Chuck1 = 0;
00014 00007          LONG Axis_IO_Chuck2 = 0;
00015 00008          LONG Axis_IO_B_Chuck2 = 0
00016      END_VAR;
00017
00018 00009          MW00061 = MW02782                " Axis Index Copy
00019
00020 00010          UFC GET_TYPE MW02782,,DB0000F DB00008 DW00001 MW00060 MW00062 MW00063;
00021
00022 00011          Axis_IDX1 = MW00061 - 1;
00023 00012          Axis_IDX2 = MW00062 - 1;
00024 00013          Axis_IDX3 = MW00063 - 1;
00025
00026 00014          MB102100[Axis_IDX1] = 1          " ABS Search Start
00027 00015          IF MW00060 == 2;
00028 00016              MB102100[Axis_IDX2] = 1      " ABS Search Start
00029 00017          IEND;
00030 00018          IF MW00060 == 3;
00031 00019              MB102100[Axis_IDX2] = 1      " ABS Search Start
00032 00020              MB102100[Axis_IDX3] = 1      " ABS Search Star
00033 00021          IEND;
00034
00035 00022          MB101200[Axis_IDX1] = 1          " While Zero Returning
00036 00023          IF MW00060 == 2;
00037 00024              MB101200[Axis_IDX2] = 1      " While Zero Returning
00038 00025          IEND;
00039 00026          IF MW00060 == 3;
00040 00027              MB101200[Axis_IDX2] = 1      " While Zero Returning
00041 00028              MB101200[Axis_IDX3] = 1      " While Zero Returning
00042 00029          IEND;
00043
00044 00030          SFORK   DW00001 == 0 ? Label0,      " Address Reset
00045                      DW00001 == 1 ? Label1,      " Sensor Search
00046                      DW00001 == 2 ? Label2,      " Sensor Search(GantryType)
00047                      DW00001 == 3 ? Label3,      " PERR Search (1 Direction)
00048                      DW00001 == 4 ? Label4,      " PERR Search (2 Direction)
00049                      DW00001 == 5 ? Label5,      " Torque Search (1 Direction)
00050                      DW00001 == 6 ? Label6,      " Torque Search (2 Direction)
00051                      DW00001 == 7 ? Label7,      " Torque Search (Shift Axis)
00052                      DW00001 == 8 ? Label8,      " Torque Search (IAI)
00053                      DEFAULT ? LabelD;
00054
00055 00031          Label0:
00056                      " Address Reset
00057
00058 00032          MSEE MPS114;                      " Abs Origin Reset
00059 00033          MB102400[Axis_IDX1] = 1          " ABS Offset 0reset Ind
00060 00034          IF MW00060 == 2;
00061 00035              MB102400[Axis_IDX2] = 1      " ABS Offset 0reset Ind
00062 00036          IEND;
00063 00037          IOW MB02700F == 1                    " Wait Address 0Reset
00064
00065 00038          JOINTO LabelX;
00066
00067 00039          Label1:
00068                      " Sensor Search
00069
00070 00040          L1_PW = (MW02782-1) * 10C          " Parameter Index
00071 00041          L1_PL = (MW02782-1) * 100 / 2    " Parameter Index
00072
00073          " SoftLimit Break
00074 00042          ML00066 = 2147483647;
00075 00043          ML00068 = -2147483648;
00076 00044          MSEE MPS115;
00077
00078          " High Speed Search
00079 00045          MB026002 = DB00008                " Moving direction(JOG/STEP)
00080 00046          ML02612 = ML40008[L1_PL]          " Speed reference setting
00081
00082 00047          MW02610 = 7;                        " Motion command <- STEP COMMAND
00083 00048          IOW (MB027010 | MB027011) == 1      " ABS ORG Input or OT Sensor
00084 00049          MW02610 = 0;                        " Motion command <- 0 Clear
00085 00050          IOW MB027009 == 1                    " Motion command response code is 0
00086

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00087 00051      TIM T100;
00088
00089      " OT센서 감지시 원점센서 감지될때까지 반대로 이동
00090 00052      IF MB027011 == 1
00091 00053          MB026002 = !DB00008
00092 00054          ML02612 = ML40008[L1_PL]
00093 00055          MW02610 = 7;
00094
00095 00056          IOW MB027010 == 1
00096 00057          MW02610 = 0;
00097 00058          IOW MB027009 == 1
00098
00099 00059      TIM T30;
00100
00101 00060          MB026002 = !DB00008
00102 00061          ML02612 = ML40008[L1_PL]
00103
00104 00062          MW02610 = 7;
00105 00063          IOW MB027010 == 0
00106 00064          MW02610 = 0;
00107 00065          IOW MB027009 == 1
00108
00109 00066      TIM T30;
00110
00111 00067          MB026002 = DB00008
00112 00068          ML02612 = ML40008[L1_PL]
00113 00069          MW02610 = 7;
00114
00115 00070          IOW MB027010 == 1
00116 00071          MW02610 = 0;
00117 00072          IOW MB027009 == 1
00118 00073      IEND;
00119
00120      " Low Speed Search
00121 00074      MB026002 = !DB00008
00122 00075      ML02612 = ML40010[L1_PL]
00123
00124 00076          MW02610 = 7;
00125 00077          IOW MB027010 == 0;
00126 00078          MW02610 = 0;
00127 00079          IOW MB027009 == 1
00128
00129 00080      TIM T100;
00130
00131      " Step Search
00132 00081      MB026002 = DB00008
00133 00082      ML02612 = ML40010[L1_PL]
00134 00083      ML02624 = 3;
00135
00136 00084      DB00000E = 0;
00137 00085      WHILE DB00000E == 0;
00138 00086          IF MB027010 == 1;
00139 00087              DB00000E = 1;
00140 00088          ELSE;
00141 00089              MW02610 = 8;
00142 00090              IOW MB02700B == 1
00143 00091              IOW MB027008 == 1
00144 00092              MW02610 = 0;
00145 00093              IOW MB027009 == 1
00146 00094          IEND;
00147 00095          EOX;
00148 00096      WEND;
00149
00150 00097      MSEE MPS114;
00151
00152 00098      MB102400[Axis_IDX1] = 1
00153 00099      IF MW00060 == 2
00154 00100          MB102400[Axis_IDX2] = 1
00155 00101      IEND;
00156
00157 00102      IF MW00060 == 3
00158 00103          MB102400[Axis_IDX2] = 1
00159 00104          MB102400[Axis_IDX3] = 1
00160 00105      IEND;
00161
00162      " SoftLimit Update
00163 00106      ML00066 = ML40094[L1_PL];
00164 00107      ML00068 = ML40098[L1_PL];
00165 00108      MSEE MPS115;
00166
00167 00109      JOINTO LabelX;
00168
00169 00110      Label2:
00170      " Sensor Search (GantryType)
00171
00172 00111      L1_PW = (MW02782-1) * 10C
00173 00112      L1_PL = (MW02782-1) * 100 / 2

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```

" OT센서
" Moving direction(JOG/STEP)
" Speed reference setting
" Motion command <- STEP COMMAND

" ABS ORG Input or OT Sensor
" Motion command <- 0 Clear
" Motion command response code is 0

" Moving direction(JOG/STEP)
" Speed reference setting

" Motion command <- STEP COMMAND
" ABS ORG Input or OT Sensor
" Motion command <- 0 Clear
" Motion command response code is 0

" Moving direction(JOG/STEP)
" Speed reference setting
" Motion command <- STEP COMMAND

" ABS ORG Input or OT Sensor
" Motion command <- 0 Clear
" Motion command response code is 0

" Moving direction(JOG/STEP)
" Speed reference setting
" Motion command <- STEP COMMAND

" Motion command <- 0 Clear
" Motion command response code is 0

" Moving direction(JOG/STEP)
" Speed reference setting
" Step travel distance

" Motion command <- STEP COMMAND
" Motion command response code not 0
" Motion command execution completed
" Motion command <- 0 Clear
" Motion command response code is 0

" Abs Origin Reset

" ABS Offset 0reset Ind
" 2 Axis Gentry
" ABS Offset 0reset Ind

" 3 Axis Gentry
" ABS Offset 0reset Ind
" ABS Offset 0reset In

" Parameter Index
" Parameter Index

```

```

00174
00175      " SoftLimit Break
00176 00113      ML00066 = 2147483647;
00177 00114      ML00068 = -2147483648;
00178 00115      MSEE MPS115;
00179
00180      " High Speed Search
00181 00116      MB026002 = DB00008      " Moving directi
on(JOG/STEP)
00182 00117      ML02612 = ML40008[L1_PL]      " Speed referenc
e setting
00183
00184 00118      MW02610 = 7;      " Motion command
<- STEP COMMAND
00185 00119      IOW (MB027010 | MB027011 | MB027017 | MB027018 | MB027019 | MB02701A) == " ABS ORG Input
or OT Sensor
00186 00120      MW02610 = 0;      " Motion command
<- 0 Clear
00187 00121      IOW MB027009 == 1      " Motion command
response code is
00188
00189 00122      TIM T100;
00190
00191      " OT센서 감지시 원점센서 감지될때까지 반대로 이동
00192 00123      IF (MB027011 | MB027018 | MB02701A) ==      " OT Sensor On Check
" High Speed Search
00193      MB026002 = !DB00008      " Moving direction(JOG/STEP)
ML02612 = ML40008[L1_PL]      " Speed reference setting
00194 00124      ML02624 = 40000      " Step travel distance
00195 00125
00196 00126      MW02610 = 8;      " Motion command <- STEP COMMAND
IOW MB02700B == 1      " Motion command response code not 0
00200 00129      IOW MB027008 == 1      " Motion command execution completed
MW02610 = 0;      " Motion command <- 0 Clear
00201 00130      IOW MB027009 == 1      " Motion command response code is 0
00202 00131
00203 00132      IEND;
00204
00205 00133      MB02600F = 1;      " ABS 실행중(Gantry)
00206 00134      PFORK Line1 Line2 Line3;
00207
00208      Line1:
00209      " High Speed Search
00210 00135      MB026002 = DB00008      " Moving direction(JOG/STEP)
00211 00136      ML02612 = ML40008[L1_PL]      " Speed reference setting
00212
00213 00137      MW02610 = 7;      " Motion command <- STEP COMMAND
00214 00138      IOW MB027010 == 1      " ABS ORG Input
00215 00139      MW02610 = 0;      " Motion command <- 0 Clear
00216 00140      IOW MB027009 == 1      " Motion command response code is 0
00217
00218 00141      JOINTO LabelX1;
00219      Line2:
00220      " High Speed Search
00221 00142      MB020002 = DB00008      " Moving direction(JOG/STEP)
00222 00143      ML02012 = ML40008[L1_PL]      " Speed reference setting
00223
00224 00144      MW02010 = 7;      " Motion command <- STEP COMMAND
00225 00145      IOW MB027017 == 1      " ABS ORG Input
00226 00146      MW02010 = 0;      " Motion command <- 0 Clear
00227 00147      IOW MB021009 == 1      " Motion command response code is 0
00228
00229 00148      JOINTO LabelX1;
00230
00231      Line3:
00232      " High Speed Search
00233 00149      MB022002 = DB00008      " Moving direction(JOG/STEP)
00234 00150      ML02212 = ML40008[L1_PL]      " Speed reference setting
00235
00236 00151      MW02210 = 7;      " Motion command <- STEP COMMAND
00237 00152      IOW MB027019 == 1      " ABS ORG Input
00238 00153      MW02210 = 0;      " Motion command <- 0 Clear
00239 00154      IOW MB023009 == 1      " Motion command response code is 0
00240
00241 00155      JOINTO LabelX1;
00242
00243 00156      LabelX1: PJOINT
00244 00157      MB02600F = 0;      " ABS 실행중(Gantry)
00245
00246      " Low Speed Search
00247 00158      MB026002 = !DB00008      " Moving direction(JOG/STEP)
00248 00159      ML02612 = ML40010[L1_PL]      " Speed reference setting
00249
00250 00160      MW02610 = 7;      " Motion command <- STEP COMMAND
00251 00161      IOW (MB027010 | MB027017 | MB027019) ==      " ABS ORG Input

```

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00252 00162      MW02610 = 0;
00253 00163      IOW MB027009 == 1
00254
00255 00164      TIM T100;
00256
00257 00165      MB02600F = 1;
00258 00166      PFORK Line4 Line5 Line6;
00259
00260      Line4:
00261          " Step Search
00262 00167      MB026002 = DB00008
00263 00168      ML02612 = ML40010[L1_PL]
00264 00169      ML02624 = 3;
00265
00266 00170      DB00000E = 0;
00267 00171      WHILE DB00000E == 0;
00268 00172          IF MB027010 == 1
00269 00173              DB00000E = 1;
00270 00174      ELSE;
00271 00175          EOX;
00272 00176          MW02610 = 8;
00273 00177          IOW MB02700B == 1
00274 00178          IOW MB027008 == 1
00275 00179          MW02610 = 0;
00276 00180          IOW MB027009 == 1
00277 00181      IEND;
00278 00182      EOX;
00279 00183      WEND;
00280
00281 00184      JOINTO LabelX2;
00282
00283      Line5:
00284          " Step Search
00285 00185      MB020002 = DB00008
00286 00186      ML02012 = ML40010[L1_PL]
00287 00187      ML02024 = 3;
00288
00289 00188      DB00000D = 0;
00290 00189      WHILE DB00000D == 0;
00291 00190          IF MB027017 == 1
00292 00191              DB00000D = 1;
00293 00192      ELSE;
00294 00193          EOX;
00295 00194          MW02010 = 8;
00296 00195          IOW MB02100B == 1
00297 00196          IOW MB021008 == 1
00298 00197          MW02010 = 0;
00299 00198          IOW MB021009 == 1
00300 00199      IEND;
00301 00200      EOX;
00302 00201      WEND;
00303
00304 00202      JOINTO LabelX2;
00305
00306      Line6:
00307          " Step Search
00308 00203      MB022002 = DB00008
00309 00204      ML02212 = ML40010[L1_PL]
00310 00205      ML02224 = 3;
00311
00312 00206      DB00000A = 0;
00313 00207      WHILE DB00000A == 0;
00314 00208          IF MB027019 == 1
00315 00209              DB00000A = 1;
00316 00210      ELSE;
00317 00211          EOX;
00318 00212          MW02210 = 8;
00319 00213          IOW MB02300B == 1
00320 00214          IOW MB023008 == 1
00321 00215          MW02210 = 0;
00322 00216          IOW MB023009 == 1
00323 00217      IEND;
00324 00218      EOX;
00325 00219      WEND;
00326
00327 00220      JOINTO LabelX2;
00328
00329 00221      LabelX2: PJOINT;
00330 00222      MB02600F = 0;
00331
00332 00223      MSEE MPS114;
00333
00334 00224      MB102400[Axis_IDX1] = 1
00335 00225      IF MW00060 == 2
00336 00226          MB102400[Axis_IDX2] = 1
00337 00227      IEND;
00338
" Motion command <- 0 Clear
" Motion command response code is 0

" ABS 실행중(Gantry)

" Moving direction(JOG/STEP)
" Speed reference setting
" Step travel distance

" ABS ORG Input

" Motion command <- STEP COMMAND
" Motion command response code not 0
" Motion command execution completed
" Motion command <- 0 Clear
" Motion command response code is 0

" Moving direction(JOG/STEP)
" Speed reference setting
" Step travel distance

" ABS ORG Input

" Motion command <- STEP COMMAND
" Motion command response code not 0
" Motion command execution completed
" Motion command <- 0 Clear
" Motion command response code is 0

" Moving direction(JOG/STEP)
" Speed reference setting
" Step travel distance

" ABS ORG Input

" Motion command <- STEP COMMAND
" Motion command response code not 0
" Motion command execution completed
" Motion command <- 0 Clear
" Motion command response code is 0

" ABS 실행중(Gantry)

" Abs Origin Reset

" ABS Offset 0reset Ind
" 2 Axis Gentry
" ABS Offset 0reset Ind

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```

00339 00228      IF MW00060 == 3                                " 3 Axis Gentry
00340 00229          MB102400[Axis_IDX2] = 1                " ABS Offset 0reset Ind
00341 00230          MB102400[Axis_IDX3] = 1                " ABS Offset 0reset In
00342 00231      IEND;
00343
00344          " SoftLimit Update
00345 00232      ML00066 = ML40094[L1_PL];
00346 00233      ML00068 = ML40098[L1_PL];
00347 00234      MSEE MPS115;
00348
00349 00235      JOINTO LabelX;
00350
00351 00236      Label3:
00352          " PERR Search (1 Direction)
00353
00354 00237      MW00060 = 1;
00355 00238      L1_PW = (MW02782-1) * 10C                        " Parameter Index
00356 00239      L1_PL = (MW02782-1) * 100 / 2                " Parameter Index
00357
00358          " SoftLimit Break
00359 00240      ML00066 = 2147483647;
00360 00241      ML00068 = -2147483648;
00361 00242      MSEE MPS115;
00362
00363 00243      IF MB000181 == 1; "SCT Use
00364 00244          OLA114 = 11000,                                " CZ
00365 00245      IEND;
00366
00367 00246      IF MB000182 | MB000183 == 1; "CCT USE
00368 00247          OLA114 = 11000,                                " LCZ
00369 00248          OLB114 = 11000,                                " RCZ
00370 00249      IEND;
00371
00372          " Position reference setting
00373 00250      ML02612 = ML40010[L1_PL]                            " Speed reference setting
00374 00251      ML02614 = ML02712 - 1000000;
00375
00376 00252      MB026005 = 1;                                    " Position refarence type <- Absplut
e Value
00377 00253      MW02610 = 1;                                    " Motion command <- STEP COMMAND
00378 00254      IOW MB027012 == 1                                " PERR 200um OVER
00379 00255      ML02614 = ML02712;                                " Calibration PERR
00380 00256      IOW MB027007 == 1                                " Motion command execution completed
00381 00257      MB026005 = 0;                                    " Position refarence type <- Increme
ntal Value
00382 00258      MW02610 = 0;                                    " Motion command <- 0 Clear
00383 00259      IOW MB027009 == 1                                " Motion command response code is 0
00384
00385 00260      MSEE MPS114;                                    " Abs Origin Reset
00386
00387          " SoftLimit Update
00388 00261      ML00066 = ML40094[L1_PL];
00389 00262      ML00068 = ML40098[L1_PL];
00390 00263      MSEE MPS115;
00391
00392 00264      TIM T100;
00393
00394 00265      MB026002 = !DB00008                                " Moving direction(JOG/STEP)
00395 00266      ML02612 = ML40012[L1_PL]                            " Speed reference setting
00396 00267      ML02624 = ML40030[L1_PL]                            " Step travel distance
00397
00398 00268      MW02610 = 8;                                    " Motion command <- STEP COMMAND
00399 00269      IOW MB02700B == 1                                " Motion command response code not 0
00400 00270      IOW MB027008 == 1                                " Motion command execution completed
00401 00271      MW02610 = 0;                                    " Motion command <- 0 Clear
00402 00272      IOW MB027009 == 1                                " Motion command response code is 0
00403
00404 00273      JOINTO LabelX;
00405
00406 00274      Label4:
00407          " PERR Search (2 Direction)
00408
00409 00275      MW00060 = 1;
00410 00276      L1_PW = (MW02782-1) * 10C                        " Parameter Index
00411 00277      L1_PL = (MW02782-1) * 100 / 2                " Parameter Index
00412
00413          " SoftLimit Break
00414 00278      ML00066 = 2147483647;
00415 00279      ML00068 = -2147483648;
00416 00280      MSEE MPS115;
00417
00418          " Position reference setting
00419 00281      ML02612 = ML40010[L1_PL]                            " Speed reference setting
00420 00282      ML02614 = ML02712 - 1000000
00421
00422 00283      MB026005 = 1;                                    " Position refarence type <- Absplut
e Value

```

```

00423 00284      MW02610 = 1;
00424 00285      IOW MB027012 == 1
00425 00286      ML02614 = ML02712;
00426 00287      IOW MB027007 == 1
00427 00288      MB026005 = 0;
      ntol Value
00428 00289      MW02610 = 0;
00429 00290      IOW MB027009 == 1
00430 00291      DL00020 = ML02712
00431
00432      " Position reference settir
00433 00292      ML02612 = ML40010[L1_PL]
00434 00293      ML02614 = ML02712 + 1000000;
00435
00436 00294      MB026005 = 1;
      e Value
00437 00295      MW02610 = 1;
00438 00296      IOW MB027012 == 1
00439 00297      ML02614 = ML02712;
00440 00298      IOW MB027007 == 1
00441 00299      MB026005 = 0;
      ntol Value
00442 00300      MW02610 = 0;
00443 00301      IOW MB027009 == 1
00444 00302      DL00022 = ML02712
00445
00446      " Move Center
00447 00303      ML02614 = (DL00020 + DL00022)/2;
00448 00304      MB026005 = 1;
      e Value
00449 00305      MW02610 = 1;
00450 00306      IOW MB027007 == 0
00451 00307      IOW MB027007 == 1
00452 00308      MB026005 = 0;
      ntol Value
00453 00309      MW02610 = 0;
00454 00310      IOW MB027009 == 1
00455
00456 00311      MSEE MPS114;
00457
00458      " SoftLimit Update
00459 00312      ML00066 = ML40094[L1_PL];
00460 00313      ML00068 = ML40098[L1_PL];
00461 00314      MSEE MPS115;
00462
00463 00315      JOINTO LabelX;
00464
00465 00316      Label5:
00466      " Torque Search (1 Direction)
00467
00468 00317      MW00060 = 1;
00469 00318      L1_PW = (MW02782-1) * 10C
00470 00319      L1_PL = (MW02782-1) * 100 / 2
00471
00472      " SoftLimit Break
00473 00320      ML00066 = 2147483647;
00474 00321      ML00068 = -2147483648;
00475 00322      MSEE MPS115;
00476
00477      " Position reference setting
00478 00323      ML02612 = ML40010[L1_PL]
00479 00324      ML02614 = ML02712 - 1000000;
00480
00481 00325      MB026005 = 1;
      e Value
00482 00326      MW02610 = 1;
00483 00327      IOW MB027013 == 1
00484 00328      ML02614 = ML02712;
00485 00329      IOW MB027007 == 1
00486 00330      MB026005 = 0;
      ntol Value
00487 00331      MW02610 = 0;
00488 00332      IOW MB027009 == 1
00489 00333      DL00020 = ML02712
00490
00491 00334      MSEE MPS114;
00492
00493      " SoftLimit Update
00494 00335      ML00066 = ML40094[L1_PL];
00495 00336      ML00068 = ML40098[L1_PL];
00496 00337      MSEE MPS115;
00497
00498 00338      TIM T100;
00499
00500 00339      MB026002 = !DB00008
00501 00340      ML02612 = ML40012[L1_PL]
00502 00341      ML02624 = ML40030[L1_PL]

```

" Motion command <- STEP COMMAND
" PERR 200um OVER
" Calibration PERR
" Motion command execution completed
" Position reference type <- Increme
" Motion command <- 0 Clear
" Motion command response code is 0
" Save - Direction Limit Pos
" Speed reference setting
" Position reference type <- Absplut
" Motion command <- STEP COMMAND
" PERR 200um OVER
" Calibration PERR
" Motion command execution completed
" Position reference type <- Increme
" Motion command <- 0 Clear
" Motion command response code is 0
" Save + Direction Limit Pos
" Position reference type <- Absplut
" Motion command <- STEP COMMAND
" Motion command execution completed
" Motion command execution completed
" Position reference type <- Increme
" Motion command <- 0 Clear
" Motion command response code is 0
" Abs Origin Reset
" Parameter Index
" Parameter Index
" Speed reference setting
" Position reference type <- Absplut
" Motion command <- STEP COMMAND
" Torque Over
" Calibration PERR
" Motion command execution completed
" Position reference type <- Increme
" Motion command <- 0 Clear
" Motion command response code is 0
" Save - Direction Limit Pos
" Abs Origin Reset
" Moving direction(JOG/STEP)
" Speed reference setting
" Step travel distance

```

00503
00504 00342      MW02610 = 8;
00505 00343      IOW MB02700B == 1
00506 00344      IOW MB027008 == 1
00507 00345      MW02610 = 0;
00508 00346      IOW MB027009 == 1
00509
00510 00347      JOINTO LabelX;
00511
00512 00348      Label6:
00513      " Torque Search (2 Direction)
00514
00515 00349      MW00060 = 1;
00516 00350      L1_PW = (MW02782-1) * 10C
00517 00351      L1_PL = (MW02782-1) * 100 / 2
00518
00519      " SoftLimit Break
00520 00352      ML00066 = 2147483647;
00521 00353      ML00068 = -2147483648;
00522 00354      MSEE MPS115;
00523
00524      " Position reference setting
00525 00355      ML02612 = ML40010[L1_PL]
00526 00356      ML02614 = ML02712 - 1000000;
00527
00528 00357      MB026005 = 1;
00529 00358      e Value
00530 00359      MW02610 = 1;
00531 00360      IOW MB027013 == 1
00532 00361      ML02614 = ML02712 + 30;
00533 00362      IOW MB027007 == 1
00534 00363      MB026005 = 0;
00535 00364      ntal Value
00536 00365      MW02610 = 0;
00537      IOW MB027009 == 1
00538
00539 00366      DL00020 = ML02712
00540 00367      " Position reference setting
00541      ML02612 = ML40010[L1_PL]
00542 00368      ML02614 = ML02712 + 1000000;
00543      e Value
00544 00369      MW02610 = 1;
00545 00370      IOW MB027013 == 1
00546 00371      ML02614 = ML02712 - ML02714
00547 00372      IOW MB027007 == 1
00548 00373      MB026005 = 0;
00549 00374      ntal Value
00550 00375      MW02610 = 0;
00551      IOW MB027009 == 1
00552
00553 00376      DL00022 = ML02712
00554 00377      " Move Center
00555 00378      ML02614 = (DL00020 + DL00022)/2;
00556 00379      MB026005 = 1;
00557 00380      e Value
00558 00381      MW02610 = 1;
00559 00382      IOW MB027007 == 0
00560 00383      IOW MB027007 == 1
00561 00384      MB026005 = 0;
00562 00385      ntal Value
00563 00386      MW02610 = 0;
00564 00387      IOW MB027009 == 1
00565 00388      MSEE MPS114;
00566 00389      " SoftLimit Update
00567 00390      ML00066 = ML40094[L1_PL];
00568 00391      ML00068 = ML40098[L1_PL];
00569 00392      MSEE MPS115;
00570 00393      JOINTO LabelX;
00571 00394      Label7:
00572 00395      " Torque Search (Shift Axis>
00573
00574 00396      L1_PW = (MW02782-1) * 10C
00575 00397      L1_PL = (MW02782-1) * 100 / 2
00576
00577      " SoftLimit Break
00578 00398      ML00066 = 2147483647;
00579 00399      ML00068 = -2147483648;
00580 00400      MSEE MPS115;
00581

```

```

" Motion command <- STEP COMMAND
" Motion command response code not 0
" Motion command execution completed
" Motion command <- 0 Clear
" Motion command response code is 0

" Parameter Index
" Parameter Index

" Speed reference setting

" Position refarence type <- Absplut

" Motion command <- STEP COMMAND
" Torque Over
" Calibration PERR
" Motion command execution completed
" Position refarence type <- Increme

" Motion command <- 0 Clear
" Motion command response code is 0

" Save - Direction Limit Pos

" Speed reference setting

" Position refarence type <- Absplut

" Motion command <- STEP COMMAND
" Torque Over
" Calibration PERR
" Motion command execution completed
" Position refarence type <- Increme

" Motion command <- 0 Clear
" Motion command response code is 0

" Save + Direction Limit P )

" Position refarence type <- Absplut

" Motion command <- STEP COMMAND
" Motion command execution completed
" Motion command execution completed
" Position refarence type <- Increme

" Motion command <- 0 Clear
" Motion command response code is 0

" Abs Origin Reset

" Parameter Index
" Parameter Index

```

```

00582          " Position reference setting
00583 00396      ML02612 = ML40010[L1_PL]          " Speed reference setting
00584 00397      ML02614 = ML02712 - 50000;
00585 00398      MB026005 = 1;          " Position reference type <- Absplut
e Value
00586
00587 00399      MW02610 = 1;          " Motion command <- STEP COMMAND
00588 00400      IOW MB027013 == 1          " Torque Over
00589
00590 00401      ML02614 = ML02712;          " Calibration PERR
00591 00402      IOW MB027007 == 1          " Motion command execution completed
00592
00593 00403      MW02610 = 0;          " Motion command <- 0 Clear
00594 00404      MB026005 = 0;          " Position reference type <- Increme
ntal Value
00595 00405      IOW MB027009 == 1          " Motion command response code is 0
00596
00597 00406      TIM T100;
00598
00599      " 스위프트축 현재위치 0으로 만들기 위해 [기존ABS OFFSET 저장량 - 현재위치] (ML53920번지는 현재위치 모니터
영역)
00600      " 값을 저장해놓고 0 좌표리셋을 해야 현재좌표가 0으로 바뀜.
00601 00407      DL00080 = ML31000[Axis_IDX1] - ML53920[Axis_IDX1];
00602
00603 00408      MSEE MPS114;          " Abs Origin Reset
00604 00409      MB102200[Axis_IDX1] = 1          " ABS Search Complete
00605
00606      " 저장해뒀던 값을 [기존ABS OFFSET 저장량]에 입력. 현재좌표 0으로 변경
00607 00410      ML31000[Axis_IDX1] = DL00080;
00608
00609 00411      TIM T30;
00610
00611 00412      MB026002 = !DB00008          " Moving direction(JOG/STEP)
00612 00413      ML02612 = ML40012[L1_PL]          " Speed reference setting
00613
00614 00414      IF MB000181 == 1; "SCT Us
00615 00415          IF MW000061 == 36          " CS
00616 00416          ML02624 = ML26374 - ML26376          " 스텝량(스위프트 중심위치 + 스위프트 원점OF
FSET)
00617 00417          IEND;
00618 00418          IEND;
00619
00620 00419      IF MB000182 | MB00183 == 1; "CCT U.
00621 00420          IF MW000061 == 36          " LCS
00622 00421          ML02624 = ML26374 - ML26376          " 스텝량(스위프트 중심위치 + 스위프트 원점OF
FSET)
00623 00422          IEND;
00624 00423          IF MW000061 == 68          " RCS
00625 00424          ML02624 = ML26974 - ML26976          " 스텝량(스위프트 중심위치 + 스위프트 원점OF
FSET)
00626 00425          IEND;
00627 00426          IEND;
00628 00427      MW02610 = 8;          " Motion command <- STEP COMMAND
00629 00428      IOW MB02700B == 1          " Motion command response code not 0
00630 00429      IOW MB027008 == 1          " Motion command execution completed
00631 00430      MW02610 = 0;          " Motion command <- 0 Clear
00632 00431      IOW MB027009 == 1          " Motion command response code is 0
00633
00634 00432      TIM T50;
00635
00636      " 다시 현재좌표 리셋
00637 00433      DL00080 = ML31000[Axis_IDX1] - ML53920[Axis_IDX1];
00638
00639 00434      MSEE MPS114;          " Abs Origin Reset
00640 00435      MB102200[Axis_IDX1] = 1          " ABS Search Complet
00641
00642 00436      ML31000[Axis_IDX1] = DL00080;
00643
00644      " SoftLimit Update
00645 00437      ML00066 = ML40094[L1_PL];
00646 00438      ML00068 = ML40098[L1_PL];
00647 00439      MSEE MPS115;
00648
00649 00440      JOINTO LabelX;
00650
00651 00441      Label18:
00652          " Torque Search (IAI
00653 00442          PFORK Line7 Line8;
00654
00655          Line7:
00656
00657 00443          Axis_IO_Chuck1 = (Axis_IDX1 - 96) * 12
00658 00444          Axis_IO_B_Chuck1 = Axis_IO_Chuck1 * 1 ;
00659
00660 00445          OBC0061[Axis_IO_B_Chuck1] = (
00661 00446          EOX;

```



```
00662
00663 00447      OBC0061[Axis_IO_B_Chuck1] = 1
00664 00448      EOX;
00665 00449      IOW IBC0299[Axis_IO_B_Chuck1] == 1
00666 00450      EOX;
00667
00668 00451      TIM T300;
00669
00670 00452      OBC0061[Axis_IO_B_Chuck1] = 0;
00671
00672 00453      JOINTO LabelX3;
00673
00674      Line8:
00675
00676 00454      IF MW00060 == 2;
00677 00455          Axis_IO_Chuck2 = (Axis_IDX2 - 96) * 12
00678 00456          Axis_IO_B_Chuck2 = Axis_IO_Chuck2 * 1 ;
00679
00680 00457      OBC0061[Axis_IO_B_Chuck2] = (
00681 00458      EOX;
00682
00683 00459      OBC0061[Axis_IO_B_Chuck2] = 1
00684 00460      EOX;
00685 00461      IOW IBC0299[Axis_IO_B_Chuck2] == 1
00686 00462      EOX;
00687
00688 00463      TIM T300;
00689
00690 00464      OBC0061[Axis_IO_B_Chuck2] = 0;
00691 00465      IEND;
00692
00693 00466      JOINTO LabelX3;
00694 00467      LabelX3: PJOINT;
00695
00696 00468      TIM T300;
00697
00698 00469      MSEE MPS114;                                " ABS Origin Reset
00699
00700 00470      JOINTO LabelX;
00701
00702 00471      LabelD:
00703 00472          JOINTO LabelX;
00704 00473      LabelX: SJOINT;
00705
00706 00474      MB102200[Axis_IDX1] = 1                                " ABS Search Complet
00707 00475      IF MW00060 == 2                                " Type : 2 Axis Gentry ABS Search
00708 00476          MB102200[Axis_IDX2] = 1                                " ABS Search Complete
00709 00477      IEND;
00710 00478      IF MW00060 == 3                                " Type : 3 Axis Gentry ABS Search
00711 00479          MB102200[Axis_IDX2] = 1                                " ABS Search Complet
00712 00480          MB102200[Axis_IDX3] = 1                                " ABS Search Complete
00713 00481      IEND;
00714
00715 00482      RET;
```